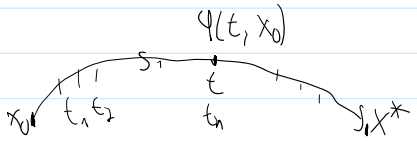
Lecture 7 - 04/06

Saturday, June 6, 2026 2:40 PM

limiting Point

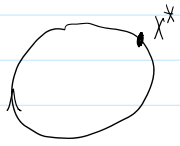


we have seen trajectories that approach the FP x^* (either a stable point, or a saddle, where there is still a few such trajetories

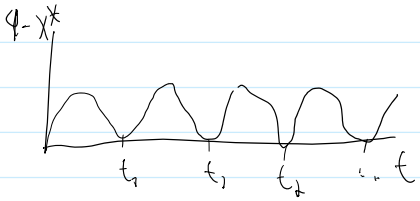
taking a point on the trajectory $\varphi(t, x_0)$.
we can discretize the trajectory to $\{t_n\}_{n\to\infty}$
so $\varphi(t_n, x_0) \to x^*$

for a limiting point $\exists \{t_n\}$ an infinit set of time points so $\varphi(t_n, x_0) \to x^*$
but it doesnt mean its true for all $\{t_n\}$

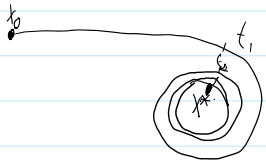
simple example - limit cycle



take a point on it
the distance from x^* ($\varphi - x^*$) changes



taking $t_1, t_2, \ldots$ is a set of infinite points where $\varphi - x^* = 0$, so, $\varphi(t_n, x_0) \to x^*$
but between these times, it can move away



another case can be -
trajectory just gets closer to the cycle, but never reaches
so any point on it is a limiting point

so x^* is a ω-limit point of x_0
All ω-limiting points are called the ω set of x_0
in reverse time, we'd call it α-limit point

example -

$\dot{x} = y$
$\dot{y} = x - 2x^3 + y(x^2 - x^4 - y^2)$

$L(x,y) = \frac{y^2}{2} + \frac{-x^2 + x^4}{2}$

$\dot{L} = y\dot{y} - x\dot{x} + 2x^3\dot{x} = y\left(x - 2x^3 + y(x^2 - x^4 - y^4)\right) + (2x^3 - x)\cdot y = y^2(-y^2 + x^2 - x^4) =$

$L = yy' - xx' + 2x^3x' = y(x - 2x^3 + y(x^2 - x^4 - y^4)) + (2x^3 - x)\cdot y = y^2(-y^2 + x^2 - x^4) =$

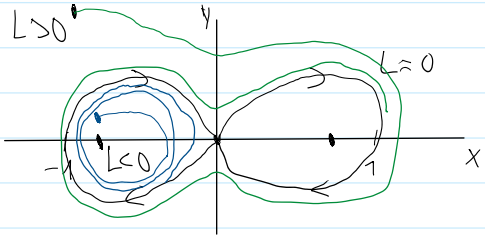
$= -2y^2 \cdot \left(\frac{y^2}{2} - \frac{-x^2 + x^4}{2}\right) = -2yL$

notice $\begin{cases} L \geq 0, & L' \leq 0 \\ L < 0, & L' \geq 0 \\ L = 0, & L' = 0 \end{cases}$

so L goes to 0, 0 is like a FP for it

$L = 0 : \quad y^2 = x^2(1 - x^2)$

$y = \pm |x| \sqrt{1 - x^2}$



so any trajectory on $L=0$ will go to 0
for any x_0 on it, 0 is the w-set of it

inside it $L<0$, outside $L>0$ and it goes towards $L=0$

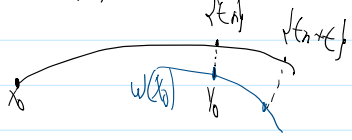
what is the w-set of x_0 ? any point on the $L=0$ figure 8
for blue x_0, w-set is left part of $L=0$

limiting set -

1) if you start on some $w(x_0)$ limiting set, you will stay on it

$y_0 \in w(x_0) \Rightarrow \varphi(t, y_0) \in w(x_0)$

$\exists \{t_n\} : \varphi(t_n, x_0) \to y_0$ we want to show that $\varphi(t, y_0) \in w(x_0)$

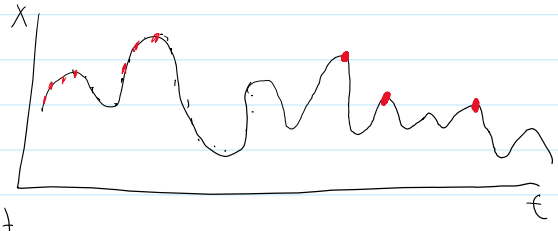


2) if a trajectory is bounded (there some area C that once you enter it, you do not leave) there must be a limiting set within it

$\varphi(t, x_0)$ is bounded $\Rightarrow w(x_0) \neq \emptyset$

proof - let's look at one dim of x

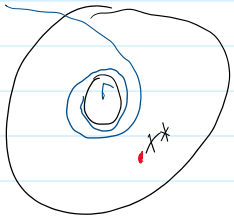
discritize it, we need to find a monotonically inc or dc set of $t, x(t)$
pick any point and continue to the next higher point
if there is none, this is a peak, we have 2 cases



pick any point and continue to the next higher point
if there is none, this is a peak, we have 2 cases
1. there is a finit number of peaks, so we can go past them, then there's just a monotonic inc set (since there are no more peaks)
2. there's an infinite set of peaks, which are decrising, so that's our monotonically dec set

now look at another dim, take the set we found as the descrite points, so now we find a subset of this subset where this direction is also dec/inc, by the same arguments, we can find it

in 2D, if a sytem is bounded, the options are



1. $\omega(x_0) = \{x^*\}$, a FP
2. $\omega(x_0) = \Gamma$, a limit cycle

so if the bounded area doesnt contain A FP, there is a cycle

example:

$$\dot{x} = y$$
$$\dot{y} = -x + y(4 - x^2 - 4y^2)$$

$L(x,y) = \frac{x^2+y^2}{2}$ L is the distance from (0,0)

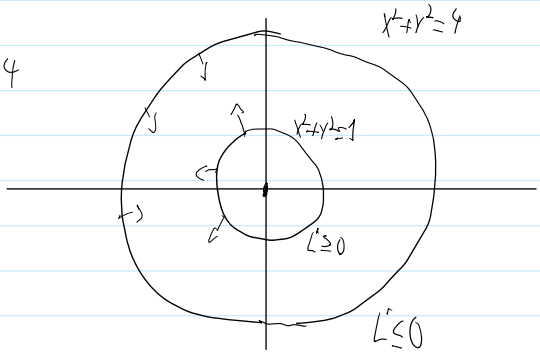
$$\dot{L} = x\dot{x} + y\dot{y} = xy + y(-x + y(4 - x^2 - 4y^2)) = y^2(4 - x^2 - 4y^2) = \begin{cases} \geq 0 & x^2+4y^2 \leq 4 \\ \leq 0 & x^2+4y^2 \geq 4 \end{cases}$$

look at unit cycle $x^2+y^2=1$

$x^2+y^2=1$

$x^2 = 1-y^2 \Rightarrow x^2+4y^2 = 1-y^2+4y^2 = 1+3y^2 \leq 4$

since $-1 \leq y \leq 1$ here



so $\dot{L} \geq 0$

look at $x^2+y^2=4$

$x^2+y^2=4$

$x^2 = 4-y^2 \Rightarrow x^2+4y^2 = 4-y^2+4y^2 = 4+3y^2 \geq 4$

so $\dot{L} \leq 0$

so the trajectories are bounded between these 2 cycles

since there are no FP within it, there is a limit cycle there